D07 · Pick-Pack-Ship & Fulfillment Execution

01 · Picking methods: single, batch, wave & zone picking

Supply Chain Reference Architecture v2.2 · mrsupplychain.ai

1 WHAT IT IS

Picking methods define how orders are collected: single-order (one order per trip), batch (several orders per trip), wave (orders released in timed groups) and zone (each picker covers one area), often combined.

2 WHY IT MATTERS

Because travel dominates picking time, **the method that minimizes travel for your order profile is one of the largest productivity levers in the warehouse (Bartholdi and Hackman — Warehouse and Distribution Science, Georgia Tech)**. Small e-commerce orders favor batching; large multi-line orders and big buildings favor zones; dispatch deadlines call for waves timed to carrier departures.

3 HOW IT IS DONE

1. **Profile orders.** Lines per order, units per line and SKU spread by channel.
2. **Match method to profile.** Batch for few-line orders, zone for large sites or many-line orders, waves around dispatch windows.
3. **Pilot before rollout.** Test batch sizes and wave timing on real volume; measure LPPH and accuracy.
4. **Re-evaluate as the mix changes.** A new channel or product range can make the current method wrong.

IN PRACTICE

Batch picking is the highest-impact improvement for warehouses with few-line orders (1–5)

In an e-commerce warehouse with 2–3 line average orders, batch picking reduces walking distance 50–65% vs. single order picking with no technology investment — just WMS configuration.

Technology-assisted picking (RF scanner, voice, pick-to-light) improves accuracy more than productivity

Picking technology (RF scanners, voice picking, put-to-light) primarily improves accuracy (eliminating human visual verification errors) and secondarily improves productivity (eliminating paper/list reading time).

Wave planning must be calibrated with carrier dispatch windows

A wave that finishes picking 2 hours before the truck departs generates operators waiting without work. A wave that finishes 30 minutes late generates orders staying in the DC until the next day. Wave planning must calibrate each wave start so picking finishes just in time for the dispatch window.

Zone picking is only efficient when the warehouse has operator density that generates congestion

Zone picking adds coordination complexity (zones must synchronize for the order to be complete) that is not justified if the warehouse doesn't have a real aisle congestion problem.

ILLUSTRATIVE CASE

Illustrative case built from typical industry values — not data from a specific company.

Illustrative case: Migration from single order picking to batch picking — e-commerce DC, 2,800 orders/day — DC processes 2,800 orders/day with a mean of 2.3 lines/order using single order picking.

Metric	Single order picking (current)	Batch picking (proposed)
LPPH (lines per hour per operator)	HIGH	48 LPPH — average walking distance: 12.4 km/operator/day

Metric	Single order picking (current)	Batch picking (proposed)
Operators required for daily volume	HIGH	38 picking operators · Labor cost: $84K/month
Optimal batch size (orders per batch)	OK	N/A -> Batch of 8 orders with WMS (item similarity + geographic zone)
Projected LPPH with batch picking	OK	Projected: 78 LPPH (+63%) · Validated in pilot: 74 LPPH (+54%)

Result: Post-implementation: LPPH from 48 to 74 (+54%). Operators required: 38 -> 25 (-13 operators). Monthly labor savings: $28K/month ($336K/year). The 3-week pilot before rollout was critical to calibrate the optimal batch size (8 orders was optimal for this DC's SKU profile).

HOW IT IS MEASURED

Pick Productivity (LPPH — Lines Picked Per Hour)

Pick Productivity (LPPH)Total picking lines completed / Total person-hours dedicated to picking

Benchmark: 60–100 LPPH for single order · 100–180 for batch/wave · 200+ for zone picking in large warehouse An LPPH drop without a volume or order-mix change indicates the picking method or slotting has deteriorated.

Pick Accuracy % (correct picking lines — correct product, correct quantity, correct location)

Pick Accuracy %(Picking lines executed without error / Total picking lines) × 100

Benchmark: >99.9% with scanner picking · >99.5% with voice picking · >98.5% for paper picking (least precise) A picking error rate of 1 in 500 lines seems small, but in a DC with 10,000 lines/day it generates 20 daily errors — each with the cost of correction, reshipping, and customer dissatisfaction.

WHAT YOU WOULD USE

Picking Technology

SAP EWM — Wave Management + Pick by Voice — Module: Wave Planning + Multi-Method PickingSAP EWM supports all picking methods (single, batch, wave, zone) with voice, RF scanner, and pick-to-light assisted picking integration.

Manhattan Associates WMS — Picking Optimization — Module: Intelligent Pick Method Selection + LPPH ManagementManhattan Associates automatically selects the optimal picking method per order based on order profile, operator availability, and dispatch windows.

Pick Assist

Honeywell Vocollect / Zebra — Module: Voice Picking + RF Scanning SolutionsThe reference voice and scanner assisted picking solutions for distribution warehouses.

4 SO WHAT

Let the order profile choose the method. Measure lines per order, match batch, zone or wave accordingly, pilot on real volume and revisit when channels change.

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